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EE/RPF

For use from May/November 2027

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Student personal code:

Extended essay - Reflection and progress form

Student: *This form is to be completed by you and your supervisor during the progress and completion of the extended essay. It records your progress through the extended essay process and includes your reflective statement addressing your extended essay experience. You must undertake three formal reflection sessions with your supervisor: The first formal reflection session should focus on your initial ideas and how you plan to undertake your research; the interim reflection session takes place once a significant amount of your research has been completed, and the final session will be in the form of a viva voce, once you have completed and handed in your extended essay. This document acts as a record in supporting the authenticity of your work. Following the viva voce, you write a* ***reflective statement*** *of no more than 500 words., which must be included on the RPF.*

The completion of this form is a mandatory requirement of the extended essay. It must be submitted together with the completed essay. The reflective statement will then be assessed under Criterion E. Students and supervisors should be aware that a mark of 0 will be awarded by the examiner for criterion E if the RPF is blank or the reflective statement is written in a language other than that of the accompanying essay.

Supervisor: *You must have three reflection sessions with each student, one early in the process, an interim meeting and then the final viva voce. Other check-in sessions are permitted but do not need to be recorded on this sheet. After each reflection session you must record the month that the reflection session took place, the student's year of DP study at that time and initial the form.*
By submitting this student work for assessment, you are taking responsibility for its authenticity. No piece of student work should be uploaded/submitted to the e-Coursework system if its authenticity is in doubt.

First reflection session

Month: DP year (1 or 2): Supervisor initials:

Interim reflection session

Month: DP year (1 or 2): Supervisor initials:

Final reflection session – Viva voce

Month: DP year (1 or 2): Supervisor initials:

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Reflective statement

(No more than 500 words, written in the same language as the accompanying extended essay)

I originally had thought of investigating a slinky and its behavior, but this attempt turned out to be quite tricky and, to create something useful for its usage, I found my topic for the essay, about the importance of designing a Stirling engine for efficiency, since this engine has shown to be a successful component in a variety of energy converters, including the Stirling Dish. The design and development of these engines for combined heat and power systems have recently come to the forefront of research and development efforts. Reading and researching about the engine helped me organize other searches that I needed for different tasks and let me to improve my time management skills.
I finally chose the research question "how do different types of Stirling engines contribute to energy generated and efficiency", because currently there are many types of Sterling engines that have been researched and improved so that there will be a new sustainable and renewable energy for humankind and reduce the usage of fossil fuels.
I discussed this with my supervisor and I found around 30 journals and papers for me to write literature reviews. Although I was worried about the lack of hands-on experience in my work, I was learning and therefore I was able to communicate these findings, something that will impact on my overall ability to communicate. I needed to use this literature review method because I could not experiment, considering that in my school there is not any Stirling dish, let alone a Stirling engine, so I used the journals for my main resources as secondary data. I decided that my topic on design and its effectiveness have allowed me to explore all of these secondary data and improve my understanding of thermodynamics.
Finally, after a long and winding road of reading every journal entry, I have completed my essay. Learning more about a topic I am interested in is always rewarding and this has certainly been the case. Several research skills, including information literacy, critical thinking, and time management have been developed by my experience with this research. Despite the problems I experienced, such as the limitation for experimentation, I was able to discover another approach, through literature reviews, through which I was able to make acceptable conclusions from my study and obtain a greater understanding of how different designs for these engines might alter their efficiency and energy output.

Word count: 400

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